সূত্র:

$$\sin\theta = \frac{1}{\operatorname{cosec}\theta}$$

$$\cos\theta = \frac{1}{\sec\theta}$$

$$\tan\theta = \frac{1}{\cot\theta} = \frac{\sin\theta}{\cos\theta} = \frac{\sec\theta}{\operatorname{cosec}\theta}$$

$$\cot\theta = \frac{1}{\tan\theta} = \frac{\cos\theta}{\sin\theta} = \frac{\operatorname{cosec}\theta}{\sin\theta}$$

$\rightarrow \sin^2\theta + \cos^2\theta = 1$

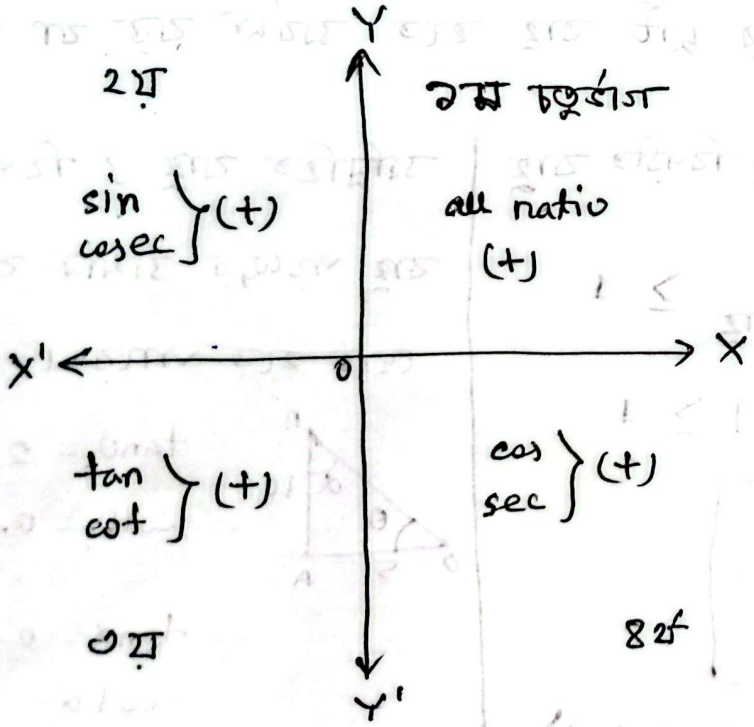
$\Rightarrow \sin^2\theta = 1 - \cos^2\theta$

$\rightarrow \sec^2\theta - \tan^2\theta = 1$

$\Rightarrow \sec^2\theta = 1 + \tan^2\theta$

$\rightarrow \operatorname{cosec}^2\theta - \cot^2\theta = 1$

$\Rightarrow \operatorname{cosec}^2\theta = 1 + \cot^2\theta$



θ / ratio	0° / 0c	30° / π/6	45° / π/4	60° / π/3	90° / π/2
sinθ	0	1/2	1/√2	√3/2	1
cosecθ	অসংজ্ঞায়িত	2	√2	2/√3	1
cosθ	1	√3/2	1/√2	1/2	0
secθ	1	2/√3	√2	2	অসংজ্ঞায়িত
tanθ	0	1/√3	1	√3	অসংজ্ঞায়িত
cotθ	অসংজ্ঞায়িত	√3	1	1/√3	0

CamScanner